WOODPECKER'S LAST KNOCK

TREND: Clear long-term trends are available for 11 species of woodpeckers, of which seven appear stable, two are declining, and two others are in rapid decline. The White-bellied woodpecker has dropped to less than 50 per cent of its historic baseline in the past three decades; the Yellow-crowned woodpecker has declined by more than 70 per cent in the past 30 years; and the Brown-capped pygmy woodpecker by 50 per cent in the long term.

WHY: Loss of large, old forest trees that woodpeckers prefer to make cavities in for nesting and fewer insects to feed on are the primary reasons for their decline.



YELLOW-CROWNED WOODPECKER

Leiopicus mahrattensis

Habitat: Forest and plantation bird found in Chhattisgarh, MP, Maharashtra, Telangana

Reason for decline: Destruction of Grassland

HIGH

SHUTTERSTOCK

Birds are a highly specialised species; fossil and biological evidence suggest they are indeed the direct descendants of dinosaurs themselves. According to research by the Simon Fraser University, the total amount of time evolutionary processes worked to create all the known bird species globally is about 77 billion years. By contrast, modern homo sapiens have had a paltry 160,000 years to get where we have. The sudden rise of humans to the top of the food chain and the simultaneous changes we have brought to the planet's weather and landscape have not given specialists like birds time to evolve and adapt. "Evolutionary biology takes millions of years to adapt to the threats most birds face today. They are specialists not just in terms of their biological requirements but also their eggs and breeding are highly sensitive. Birds, for instance, won't breed if they are traumatised," says Prabhu.

THE HUMAN THREAT

In February 2020, images of a rare Oriental darter (*Anhinga melanogaster*) in Palwal, Haryana, its beak encased in plastic, took the internet by storm. A couple of years earlier, a Black-necked stork (*Ephippiorhynchus asiaticus*) in a wetland outside Delhi stared in horror at the camera as a plastic ring sealed its beak. Both birds were eventually rescued, but not all are as lucky.

Exposure to toxic chemicals—heavy metals in pollution, plastics and organophosphates used in several insecticides—wreaks the greatest havoc on birds, be it hormone disruption, alterations in feeding behaviour or a compromised immune system. Such exposure during the reproductive stage can affect fertility, egg formation and chick-rearing behaviour. Studies have shown that toxic metals infiltrate all organs of a bird. A 2020 study in *Scientific Reports* noted unusually high quantities of toxic substances found in the feathers of 15 species of shorebirds at the Point Calimere Wildlife Sanctuary, a Ramsar site (wetland site designated to be of international importance per Ramsar Convention, 1971), and in the Pichavaram Mangrove Forest. Another 2022 study in the journal *Toxics* found the levels of metals in the tissues, liver, kidney and feathers of the Indian pond heron (*Ardeola grayii*) and the Black-crowned night heron (*Nycticorax nycticorax*) to be over permissible limits.



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– MRUGANK PRABHU *Researcher, ornithologist*